

1. Steam enters a turbine operating at steady state at 2 MPa, 360°C with a velocity of 100 m/s. Saturated vapor exits at 0.1 MPa and a velocity of 50 m/s. The elevation of the inlet is 3 m higher than at the exit. The mass flow rate of the steam is 15 kg/s, and the power developed is 7 MW. Let $g = 9.81 \text{ m/s}^2$. Determine (a) the area at the inlet, in m^2 , and (b) the rate of heat transfer between the turbine and its surroundings, in kW.